AN EXPLICIT R-FLAT COMPLEX COMPUTING RHOM BETWEEN A PSEUDO-COHERENT OBJECT AND AN OBJECT OF FINITE TOR DIMENSION, AND ITS BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 22039-22062 (canonical tag 0E1U) and the counted proof is the proof environment at source lines 22064-22102, which immediately follows it and is its own proof.

Editorial note. The printed text contains the slips since F is bounded an since P is bounded above at source line 22076, where and is meant. The text is copied exactly as it stands and is not corrected.

Editorial note. The three claims of the statement that are not proved in the span itself are the two choices of representatives and the finite presentation refinement of the last two sentences; the first is the definition of pseudo-coherence, the second is derived from bounded tor dimension together with the truncation and the cited flatness statement, and the refinement is a choice that the preceding arguments support, since a relatively perfect object admits finitely presented representatives.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- A pseudo-coherent complex is by definition represented by a bounded above complex of finite free modules, and a bounded tor dimension implies boundedness, so a bounded above complex of free modules represents the object and can be truncated. used at proof lines 22065-22072. Produces the two complexes P and F of the statement and makes F a bounded complex of R-flat modules.
- For a bounded above complex of finite free modules and a bounded complex of flat modules the Hom complex is a bounded below complex whose terms are finite direct sums of the Hom modules in the matching degrees. used at proof lines 22075-22082. Gives the termwise description of E and shows

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that its terms are R-flat, which is assertion (a) apart from representing RHom.
- Base change for RHom along R -> R' and the compatibility of Hom with base change for the terms, using flatness of the terms over R. used at proof lines 22087-22101. Identifies E tensor_R R' with the Hom complex of the base changed complexes, which is assertion (b).

Source reference. Original lemma 0653: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. The reference is used at proof line 22072 to see that the truncation of F has R-flat terms; the termwise computation of the Hom complex that follows is inside the counted span.

Source reference. Original lemma 0A66: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18708–18708. The reference identifies the complex E with RHom_A(K, L) at proof lines 22083-22084; the termwise description of E that makes it a bounded below complex of R-flat modules is inside the counted span.

Source reference. Original lemma 0661: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14955–14955. The reference is used at proof lines 22089-22092 to see that the base changed complexes represent the base changed objects; the subsequent termwise identification of E tensor_R R' with the Hom complex of those representatives is inside the counted span.

Lemma 0.2. *Let $R \to A$ be a flat ring map. Let $K, L \in D(A)$ with K pseudo-coherent and L finite tor dimension over R. We may choose*

(1) *a bounded above complex $P^\bullet$ of finite free A-modules representing K, and*
(2) *a bounded complex of R-flat A-modules $F^\bullet$ representing L.*

Given these choices we have

(a) *$E^\bullet = \mathrm{Hom}^\bullet(P^\bullet, F^\bullet)$ is a bounded below complex of R-flat A-modules representing $R\mathrm{Hom}_A(K, L)$,*
(b) *for any ring map $R \to R'$ with $A' = A \otimes_R R'$ the complex $E^\bullet \otimes_R R'$ represents $R\mathrm{Hom}_{A'}(K \otimes^{\mathbf{L}}_A A', L \otimes^{\mathbf{L}}_A A')$.*

If in addition $R \to A$ is of finite presentation and L is R-perfect, then we may choose F^p to be finitely presented A-modules and consequently E^n will be finitely presented A-modules as well.

Proof. The existence of $P^\bullet$ is the definition of a pseudo-coherent complex. We first represent L by a bounded above complex $F^\bullet$ of free A-modules (this is possible because bounded tor dimension in particular implies bounded). Next, say L viewed as an object of $D(R)$ has tor amplitude in $[a, b]$. Then, after replacing $F^\bullet$ by $\tau_{\geq a}F^\bullet$, we get a complex as in (2). This follows from Lemma 0653.

Proof of (a). Since $F^\bullet$ is bounded an since $P^\bullet$ is bounded above, we see that $E^n = 0$ for $n \ll 0$ and that E^n is a finite (!) direct sum

$$E^n = \bigoplus\nolimits_{p+q=n} \mathrm{Hom}_A(P^{-q}, F^p)$$

and since P^{-q} is finite free, this is indeed an R-flat A-module. The fact that $E^\bullet$ represents $R\mathrm{Hom}_A(K, L)$ follows from Lemma 0A66.

Proof of (b). Let $R \to R'$ be a ring map and $A' = A \otimes_R R'$. By Lemma 0661 the object $L \otimes^{\mathbf{L}}_A A'$ is represented by $F^\bullet \otimes_R R'$ viewed as a complex of A'-modules (by flatness of F^p over R). Similarly for $P^\bullet \otimes_R R'$. As above $R\mathrm{Hom}_{A'}(K \otimes^{\mathbf{L}}_A A', L \otimes^{\mathbf{L}}_A A')$ is represented by

$$\mathrm{Hom}^\bullet(P^\bullet \otimes_R R', F^\bullet \otimes_R R') = E^\bullet \otimes_R R'$$

The equality holds by looking at the terms of the complex individually and using that $\mathrm{Hom}_{A'}(P^{-q} \otimes_R R', F^p \otimes_R R') = \mathrm{Hom}_A(P^{-q}, F^p) \otimes_R R'$. □